

UbiSense 2025: First International Workshop on Emerging Techniques in Ubiquitous Sensing and Interaction

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Abstract

As wearable and ubiquitous sensing technologies become increasingly embedded in our daily environments, new opportunities and challenges are emerging around how to effectively collect, analyse, and act upon the resulting data. This creates exciting opportunities for personalized healthcare, energy optimization, smart city services, and behavioural insights. The workshop will bring together researchers, practitioners, and industry professionals to explore cutting-edge techniques in ubiquitous sensing and interaction. We will examine how recent advances in artificial intelligence, especially large language models (LLMs), multimodal models, and context-aware algorithms, can be leveraged to improve real-time data processing, cross-modal understanding, and intelligent interaction design. The event includes high-quality research papers and invited keynote talks.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing; • Applied computing;

Keywords

Large Language Models; UbiComp; Wearable Sensing

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1 Introduction and Motivation

Ubiquitous sensing has become a cornerstone of modern computing, enabling seamless interactions between humans, devices, and environments. The ubiquity of sensors and sensing technologies across diverse domains ranging from wearable devices to urban-scale IoT networks including smart homes, smart cities, healthcare, transportation, and environmental monitoring, has led to an explosion of sensor-generated data [8]. These sensors capture diverse signals, such as motion, sound, environmental conditions, and physiological states, enabling unprecedented insights into human activities [4, 5], mental wellbeing [3], environmental dynamics, and socio-technical systems [10].

Recent advancements in AI, particularly large language models (LLMs) [1, 2] and multimodal foundation models, has further expanded the potential to interpret this data, transforming raw sensor inputs into actionable knowledge. These models, renowned for their ability to generalize across domains and learn from unstructured data, offer a promising pathway to address longstanding challenges in sensor data analysis, such as handling spatio-temporal variability, noisy inputs, and cross-modal fusion [6]. Despite these promising advancements, integrating foundation models into ubiquitous sensing remains a challenging endeavor. Key obstacles include lack of well-established ubiquitous sensing data and knowledge bases, handling heterogeneous sensor data, ensuring efficient adaptation to diverse environments, addressing privacy concerns, and deploying models on resource-constrained devices [7, 11].

This workshop is motivated by the urgent need to bridge the divide between cutting-edge AI advancements and the practical realities of sensor-driven applications. This workshop, *UbiSense: Emerging Techniques in Ubiquitous Sensing and Interaction*, aims to bring together researchers and practitioners from AI, ubiquitous computing, and sensor technologies to explore these challenges and advance state-of-the-art methodologies for intelligent sensing and interaction. By fostering interdisciplinary collaboration, this workshop will help shape the future of AI-driven ubiquitous sensing systems and their real-world applications.

Relation to UbiComp: UbiComp has long been dedicated to advancing the vision of computing that seamlessly integrates into

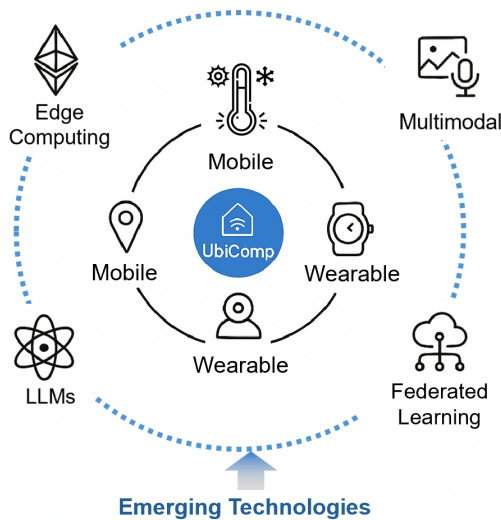


Figure 1: The scope of the UbiSense workshop.

everyday life. As sensor technologies and embedded computing become more pervasive, the role of AI in interpreting and acting upon ubiquitous sensor data is more critical than ever. This workshop aligns closely with UbiComp’s mission by **exploring how cutting-edge AI models and emerging techniques such as the recent LLMs and multimodal foundation models can enhance intelligent environments, adaptive systems, and human-computer interactions** [9]. With the increasing availability of sensor-rich environments, the demand for advanced techniques to process, analyze, and interpret sensor data in natural language that humans can easily understand continues to grow. *UbiSense* will serve as a **bridge between AI-driven advancements and the core challenges of ubiquitous computing**, including but not limited to sensing data integration, energy efficiency, privacy-preserving computation, and context-aware decision-making. By fostering discussions on integrating emerging technologies into ubiquitous sensing, this workshop will contribute to UbiComp’s broader goal of making computing more natural, intuitive, and seamlessly embedded in the physical world.

2 Objective and Scope

Our objective is to provide a platform for researchers, practitioners, and stakeholders from diverse fields to explore the unique challenges and opportunities presented by AI-driven ubiquitous sensing. As ubiquitous sensing environments generate vast and heterogeneous data streams, conventional methods struggle to scale effectively across different domains and tasks. Foundation models, trained on large-scale data and adaptable to various sensor-based applications, offer a promising paradigm shift in ubiquitous sensing. As shown in Figure 1, the workshop will focus on these key areas:

- *Adapting LLMs for Ubiquitous Sensing*: Unlike image and text data, ubiquitous sensing data exhibits unique characteristics such as continuous streaming, multi-modal and heterogeneous formats, varying sampling rates, and strong dependencies on spatial and temporal contexts. To enable LLMs to

effectively process and learn from sensor data, new strategies are required, including temporal sequence modeling, real-time adaptation, and efficient representation learning for diverse sensor modalities.

- *Multimodal Integration for Sensing Applications*: Ubiquitous sensing involves diverse data modalities, such as vision, audio, motion, and physiological signals. Effective multimodal fusion techniques are crucial for extracting meaningful patterns and improving system robustness. This topic explores strategies for integrating multiple sensor streams using AI models, enabling richer context-awareness and more reliable predictions.
- *Foundation Models for Sensor Data Learning*: Foundation models have the potential to generalize across various sensing tasks, reducing the need for extensive task-specific training. This section focuses on leveraging pre-trained foundation models for ubiquitous sensing, exploring techniques such as fine-tuning, transfer learning, and self-supervised learning to enhance performance in sensor-driven applications.
- *Datasets and Benchmarking*: Developing standardized datasets is a key enabler to support the training of sensing foundation models. Benchmarks are also essential for evaluating existing emerging AI models in ubiquitous sensing contexts. This discussion will cover the challenges of collecting, annotating, and sharing large-scale sensor datasets, as well as the need for reproducible and fair benchmarking methodologies that reflect real-world deployment scenarios.
- *Novel Applications and Interface Design*: The application of emerging AI techniques in ubiquitous sensing could foster new applications. This focus area will facilitate innovative use cases and explore human-AI interaction design, ensuring that AI-driven sensing systems provide intuitive, efficient, and privacy-aware user experiences across different environments.

3 Topics of Interest

The workshop will provide a venue for discussions on these topics:

- Adaptation of cutting-edge machine learning algorithms for modelling sensing data from different sources.
- Model architecture and tokenization approaches for sensing data.
- Advanced multi-modal learning frameworks (e.g., self-supervised learning, transfer learning, adversarial learning) for sensing data.
- Cross-domain and multi-modal data fusion, integrating spatio-temporal, visual, and textual information sourced from smart wearable devices.
- LLM-powered user interface for sensing data presentation.
- Knowledge distillation methods for distilling knowledge from LLMs to enhance domain-specific applications.
- Real-world applications of sensor data mining such as in the healthcare domain.
- Developing foundational models or utilizing LLMs for processing and analyzing data extracted from wearable sensors.
- Edge computing and low-power AI for real-time processing of sensor data.
- Human-centered interaction design informed by continuous behavioural sensing.

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